

MRPL Refinery — Standard Operating Procedure: Reciprocating Compressor C-101

Document Control

Document ID: SOP-024 | Classification: RESTRICTED | Unit: Hydrogen Recovery Unit | Revision: 4
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Equipment Specification

Parameter	Value
Equipment Tag	C-101
Equipment Type	Reciprocating Compressor, 2-Stage
Service	Hydrogen Recycle Gas Compression
Location	Hydrogen Recovery Unit, Building 4
Discharge Pressure Rating	42 bar g
Rated Power	1250 kW

Normal Operating Envelope

Parameter	Normal Range	Warning Band	Critical / Trip
Discharge Pressure	28-34 bar g	34-38 bar g	> 38 bar g
Discharge Temperature	95-120 C	120-135 C	> 135 C
Cylinder Vibration	< 4.0 mm/s	4.0-6.5 mm/s	> 6.5 mm/s
Crankcase Oil Pressure	2.5-3.5 bar g	2.0-2.5 bar g	< 2.0 bar g

Startup Sequence

1. Confirm suction and discharge block valves are in correct startup position per P&ID; HRU-014.
2. Verify lube oil pressure reaches minimum 2.5 bar g before engaging clutch.
3. Engage compressor at reduced load (40 percent) for 15 minutes; monitor discharge temperature rise.
4. Ramp to full load only after cylinder wall temperatures stabilize within normal range.

High Discharge Temperature Response

If discharge temperature exceeds 135 C: (1) Reduce load immediately to 50 percent. (2) Check interstage cooler water flow and fouling. (3) If temperature does not recede within 10 minutes, initiate controlled shutdown per SOP-024 Section 5.1 and notify Unit Engineer. High discharge temperature combined with rising vibration may indicate valve plate failure or piston ring wear, requiring internal inspection before restart.